



NPTEL Online Certification Course
Generative AI for Computer Vision, IIT Guwahati
Assignment 7: Detailed Solution
Indian Institute of Technology Guwahati



1. What is the main purpose of the attention mechanism?

- a) To focus on the most relevant parts of the input (✓)
- b) To remove all input information
- c) To reduce the number of classes
- d) To increase image size

Answer: a

Detailed Explanation: Attention allows a model to assign more importance to relevant parts of the input when producing an output.

2. In self-attention, the model mainly relates:

- a) Only the first and last elements
- b) Only the input and output images
- c) Different training datasets
- d) Different parts of the same input sequence (✓)

Answer: d

Detailed Explanation: Self-attention allows each element of a sequence to interact with other elements of the same sequence.



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3. What is the purpose of masked self-attention?

- a) To prevent a position from attending to future positions (✓)
- b) To remove all attention scores
- c) To increase the image resolution
- d) To remove the input sequence

Answer: a

Detailed Explanation: Masked self-attention prevents a position from using information from future positions, which is important when generating sequences step by step.

4. Multi-head self-attention uses:

- a) Multiple input images only
- b) Only one attention score
- c) Multiple attention heads to capture different relationships (✓)
- d) No attention mechanism

Answer: c

Detailed Explanation: Multi-head self-attention performs attention using multiple heads, allowing the model to learn different types of relationships in the input.



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5. A Transformer is mainly based on:

- a) Pooling only
- b) Attention mechanisms (✓)
- c) Decision trees
- d) K-means clustering

Answer: b

Detailed Explanation: Transformers use attention mechanisms as a central component for modelling relationships between elements in the input.

6. How does a Vision Transformer (ViT) process an image?

- a) It uses only pooling layers
- b) It converts the image directly into audio
- c) It removes all image information
- d) It divides the image into patches and processes them as tokens (✓)

Answer: d

Detailed Explanation: Vision Transformers divide an image into smaller patches, convert the patches into embeddings, and process them as a sequence of tokens.



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7. One major difference between a Transformer and a CNN is that:

- a) Transformers use attention to model relationships between input elements (✓)
- b) Transformers cannot process images
- c) CNNs do not have trainable parameters
- d) CNNs can only process text

Answer: a

Detailed Explanation: CNNs primarily use convolution operations to capture local spatial patterns, while Transformers use attention to model relationships between different input elements.

8. What is the main role of the Transformer encoder?

- a) To increase image resolution
- b) To delete the input sequence
- c) To transform the input into useful contextual representations (✓)
- d) To store training data

Answer: c

Detailed Explanation: The Transformer encoder processes the input and produces contextual representations that capture relationships among the input elements.



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9. What is the purpose of input embedding?

- a) To convert input tokens into numerical vector representations (✓)
- b) To remove the input tokens
- c) To increase the number of images
- d) To calculate the final accuracy

Answer: a

Detailed Explanation: Input embedding converts discrete input tokens, such as words or image patches, into numerical vectors that can be processed by the Transformer.

10. Why is positional encoding used in a Transformer?

- a) To remove the input sequence
- b) To provide information about the position or order of input elements (✓)
- c) To reduce the number of attention heads
- d) To increase the image resolution

Answer: b

Detailed Explanation: Unlike recurrent networks, Transformers do not inherently know the order of input elements. Positional encoding provides information about the position of each element in the sequence.